

Question number	Answer	Additional guidance	Mark
3(a)	An answer that includes two of the following points: • TB • Cystic Fibrosis • Asthma • COPD • Lung cancer • emphysema • bronchitis	examiners to check other alternatives Covid long covid pneumonia pulmonary fibrosis accept any named respiratory disease	Expert (2)

Question number	Answer	Additional guidance	Mark																				
3(b)(i)	An answer that includes three of the following points: - as body mass increases the (mean resting) ventilation rate decreases / inverse relationship / negative correlation (1) - comment on the small number of {animals / species/ sample size} (1) - correlation doesn't imply causation / other aspects affect resting ventilation rate (1) - appropriate comment on standard deviations (1)	accept converse accept significant (difference) between xxx and xx because SDs do not overlap not a significant (difference) between xxx and xxx because SDs do overlap SD values are small which indicates higher {validity/reliability} don't accept {overlapping / not overlapping} SDs are (not significant / significant) unqualified ignore not-overlapping /overlapping unqualified? <table border="1"> <tr> <td>talapoin</td><td>0.45</td><td>0.54</td><td>0.63</td></tr> <tr> <td>lemur</td><td>0.49</td><td>0.53</td><td>0.57</td></tr> <tr> <td>porcupine</td><td>0.27</td><td>0.30</td><td>0.33</td></tr> <tr> <td>tiger</td><td>0.21</td><td>0.27</td><td>0.33</td></tr> <tr> <td>hippo</td><td>0.09</td><td>0.12</td><td>0.15</td></tr> </table>	talapoin	0.45	0.54	0.63	lemur	0.49	0.53	0.57	porcupine	0.27	0.30	0.33	tiger	0.21	0.27	0.33	hippo	0.09	0.12	0.15	Expert (3)
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3(b)(ii)	A calculation showing the following steps: • calculation of number of breaths per minute (1) • calculation of tidal volume in dm³ and answer given to 2 significant figures (1)	$(0.54 \times 60) = 32.4$ $(32.4 \times 0.04) = 1.3 \text{ dm}^3$ include the other way round e.g. $0.54 \times 0.04 = 21.6$ $(21.6 \times 60) = 1.3 \text{ dm}^3$ correct answer scores full marks 1.30 = 1 mark only 12960/1296/ 1.296 / 32.4 / 21.6 / $(0.54 \times 60) / (0.54 \times 0.04) /$ $(0.54 \times 40/1000)$ somewhere = 1 mark no ecf	Graduate (2)

Question number	Answer	Additional guidance	Mark
3(c)	A description that includes four of the following points: • (because) increase {activity/ aerobic respiration} leads to an {increase in carbon dioxide /decrease in pH}(1) • which is detected by chemoreceptors (1) • (which causes transmission of) impulses to the {respiratory centre / medulla (oblongata)} (1) • which (increases) impulses (via SNS) to increase contraction of {breathing / intercostal / diaphragm} muscles (1) • increase in ventilation rate (1)	do not accept signals or messages for impulses accept impulses to ventilation centre accept impulses (via SNS) for further contraction of {breathing / intercostal / diaphragm} muscles accept increase breathing rate	Expert (4)

